

SAT Math Mastery · Week 1B**Lines, Slope & Linear Functions — Answer Key & Worked Solutions****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** slope from points/graphs/tables · forms of a line · intercepts · parallel & perpendicular · translations · meaning of slope/intercept · linear function notation**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Find slope from two points, a graph, a table, a standard-form equation, or a verbal rate.
- S2.** Write and convert between slope-intercept, point-slope, standard and intercept forms.
- S3.** Find x - and y -intercepts and use them (areas, conditions, context meaning).
- S4.** Parallel and perpendicular lines, including directly from $Ax + By = C$.
- S5.** Translate a line (up/down/left/right) and track slope and intercepts.
- S6.** Interpret slope and intercept in context; build $f(x) = mx + b$ from data and solve for inputs.
- S7.** Coordinate reasoning: collinearity, midpoint, perpendicular bisector, triangles cut by axes.
- S8.** Parameter conditions on lines (a constant fixes slope, intercept, perpendicularity or area).

2 · REQUIRED FACTS

- Slope = $\frac{\text{rise}}{\text{run}}$ = constant rate of change; the y -intercept is the value at $x = 0$; the x -intercept is where $y = 0$.
- Horizontal line $y = c$: slope 0. Vertical line $x = c$: slope undefined.
- Parallel $\iff$ equal slopes (different intercepts). Perpendicular $\iff$ slopes are negative reciprocals (product -1).
- A function is linear if equal steps in x give equal steps in y (check every gap in a table).
- Shifting a graph up k adds k to y ; slope never changes under a translation.
- Two points determine a line; on a graph read only points that sit exactly on grid intersections.

3 · REQUIRED FORMULAS

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y = mx + b, \quad y - y_1 = m(x - x_1), \quad \frac{x}{a} + \frac{y}{b} = 1$$

$$Ax + By = C: \quad m = -\frac{A}{B}, \quad x\text{-int } \frac{C}{A}, \quad y\text{-int } \frac{C}{B}, \quad \text{area with axes} = \frac{1}{2} \left| \frac{C}{A} \cdot \frac{C}{B} \right|$$

$$\parallel: A_1B_2 = A_2B_1, \quad \perp: A_1A_2 + B_1B_2 = 0, \quad \text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \quad f(a + d) - f(a) = md$$

4 · MINI THEORY SHEET**Choose the form by the data.**

Given	Fastest route
two points	$m = \frac{\Delta y}{\Delta x}$, then point-slope with either point
slope and y -intercept	$y = mx + b$ directly
both intercepts $(a, 0), (0, b)$	$\frac{x}{a} + \frac{y}{b} = 1$, or $m = -\frac{b}{a}$
$Ax + By = C$	read $m = -\frac{A}{B}$; set $x = 0$ or $y = 0$ for intercepts
a table	m from any two rows; confirm the rest have the same m

Perpendicular in standard form. Swap the coefficients and negate one: $\perp$ to $Ax + By = C$ is $Bx - Ay = C'$; find C' from the given point.

Parallel in standard form. Keep A, B (or scale both), change only C .

Translation. Down k : $b \rightarrow b - k$, and the x -intercept moves by $\frac{k}{m}$. Same slope, always.

Meaning. In $y = mx + b$: m = change in y per 1 unit of x ; b = starting value. Units of m are (units of y)/(units of x).

Constant conditions. Turn the condition (slope, perpendicular, area, collinear) into one equation in the constant, then solve.

Whole-number contexts. Solve $f(x) = \text{target}$; if x must be a whole day/trip/session, round to the first value that *reaches* the target.

Speed shortcuts.

Situation	Shortcut
equal input steps	$f(a + d) - f(a) = md$ (no equation needed)
perpendicular, standard form	$Ax + By = C \rightarrow Bx - Ay = C'$, find C' from the point
translate right h / up k	replace x by $x - h$; add k to y
triangle with axes	area = $\frac{C^2}{2 AB }$ for $Ax + By = C$
perpendicular bisector	midpoint + negative reciprocal of the segment's slope
sign of constants	$m = -\frac{A}{B}$, y -int $\frac{C}{B}$, x -int $\frac{C}{A}$: signs fix the quadrants
intersection with a parameter	solve for x first; the parameter may cancel
infinitely many / no solutions	scale one equation: same ratios $\Rightarrow$ same line

5 · GUIDED EXAMPLES

Example 1 — Two points $\rightarrow$ equation $\rightarrow$ intercepts

A line passes through $(-1, 7)$ and $(3, -1)$. Find its equation and both intercepts.

$$m = \frac{-1 - 7}{3 - (-1)} = \frac{-8}{4} = -2. \text{ Point-slope: } y - 7 = -2(x + 1) \Rightarrow y = -2x + 5.$$

$$y\text{-intercept } 5; x\text{-intercept: } 0 = -2x + 5 \Rightarrow x = \frac{5}{2}.$$

Example 2 — Perpendicular through a point (standard form)

Line ℓ : $4x - 3y = 12$. Write the line through $(2, -1)$ perpendicular to ℓ .

Swap and negate: $3x + 4y = C'$. Plug in $(2, -1)$: $C' = 3(2) + 4(-1) = 2$. Answer: $3x + 4y = 2$.

Example 3 — Meaning of slope and intercept

A gym plan costs \$340 after 2 months and \$640 after 5 months (linear). Write $C(m)$ and interpret its parts.

$$\text{slope} = \frac{640 - 340}{5 - 2} = 100. \quad C(m) = 100m + b; \quad 340 = 200 + b \Rightarrow b = 140. \quad \text{So } C(m) = 100m + 140.$$

100 = dollars per month; 140 = joining fee (cost at $m = 0$).

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 7x - 4$. What is the value of $f(3) + f(-1)$?

Worked solution

$$1. f(3) = 7(3) - 4 = 17.$$

$$2. f(-1) = 7(-1) - 4 = -11.$$

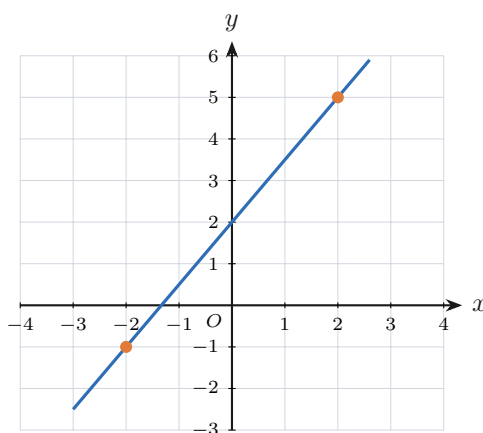
$$3. f(3) + f(-1) = 17 + (-11) = 6.$$

Answer: 6

2.

EASY · DESMOS

The graph of a line in the xy -plane is shown. Which equation represents the line?



(A) $y = -\frac{3}{2}x + 2$

(B) $y = \frac{3}{2}x + 2$ ✓

(C) $y = \frac{2}{3}x + 2$

(D) $y = \frac{3}{2}x - 2$

Worked solution

1. Two points on grid intersections: $(-2, -1)$ and $(2, 5)$.

$$2. m = \frac{5 - (-1)}{2 - (-2)} = \frac{6}{4} = \frac{3}{2}.$$

3. $5 = \frac{3}{2}(2) + b \Rightarrow b = 2$; the line crosses the y -axis at $(0, 2)$.

$$4. y = \frac{3}{2}x + 2.$$

Answer: (B)

3.

EASY · NO CALCULATOR NEEDED

What is the x -intercept of the line given by $3x + 4y = 24$?

(A) $(0, 8)$ (B) $(6, 0)$ (C) $(8, 0)$ ✓(D) $(0, 6)$ **Worked solution**

1. At an x -intercept, $y = 0$.
2. $3x = 24$, so $x = 8$.
3. The x -intercept is the point $(8, 0)$.

Answer: (C)

4.

EASY · CALCULATOR

A tank contains 90 liters of water after 2 minutes of draining and 66 liters after 6 minutes. The water level decreases at a constant rate. What is the rate of change of the volume, in liters per minute? (Enter the signed value.)

Worked solution

1. $\text{rate} = \frac{66 - 90}{6 - 2} = \frac{-24}{4}$.
2. -6 liters per minute (negative because the volume is decreasing).

Answer: -6

5.

EASY · NO CALCULATOR NEEDED

Which of the following lines is perpendicular to the line $2x + y = 7$?

(A) $x - 2y = 4$ ✓(B) $2x - y = 3$ (C) $x + 2y = 1$ (D) $2x + y = -3$ **Worked solution**

1. Slope of $2x + y = 7$: $-\frac{A}{B} = -2$.
2. A perpendicular slope is the negative reciprocal, $\frac{1}{2}$.
3. $x - 2y = 4 \Rightarrow y = \frac{1}{2}x - 2$, slope $\frac{1}{2}$. (Swap-and-negate: $2x + y \rightarrow x - 2y$.)

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

The function f is linear. Selected values are shown in the table. What is the value of p ?

x	$f(x)$
1	7
4	19
6	p

(A) 31

(B) 21

(C) 23

(D) 27 ✓

Worked solution

1. Slope from the first two rows: $\frac{19 - 7}{4 - 1} = 4$.

2. From $x = 4$ to $x = 6$ the input rises by 2, so f rises by $4 \cdot 2 = 8$.

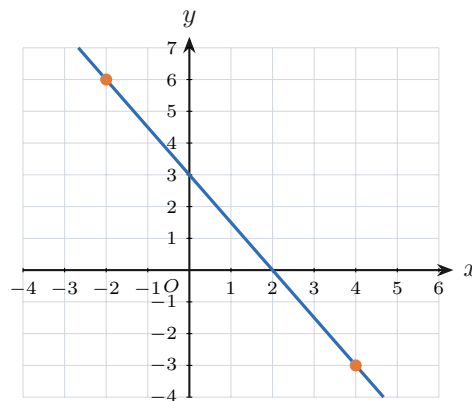
3. $p = 19 + 8 = 27$.

Answer: (D)

7.

MEDIUM · DESMOS

The graph of a line is shown. Which equation represents it?

(A) $3x - 2y = 6$ (B) $3x + 2y = -6$ (C) $3x + 2y = 6$ ✓(D) $2x + 3y = 6$ **Worked solution**

1. Read $(-2, 6)$ and $(4, -3)$.

2. Test each option at both points. $3x + 2y = 6$: $3(-2) + 2(6) = 6$ ✓ and $3(4) + 2(-3) = 6$ ✓.

3. The others fail at least one point.

Answer: (C)

8.

MEDIUM · NO CALCULATOR NEEDED

Which of the following is an equation of the line that passes through $(-4, 3)$ and is perpendicular to the line $5x - 2y = 9$?

(A) $5x + 2y = -14$

(B) $2x + 5y = 7$ ✓

(C) $2x - 5y = -23$

(D) $5x - 2y = -26$

Worked solution

1. Swap and negate: a perpendicular to $5x - 2y = C$ has the form $2x + 5y = C'$.
2. Substitute $(-4, 3)$: $C' = 2(-4) + 5(3) = 7$.
3. $2x + 5y = 7$.

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

The line $y = -2x + 6$ is translated 3 units to the right. Which equation defines the new line?

(A) $y = -2x$

(B) $y = -2x + 9$

(C) $y = -2x + 3$

(D) $y = -2x + 12$ ✓

Worked solution

1. Moving right by 3 replaces x with $x - 3$: $y = -2(x - 3) + 6$.
2. $y = -2x + 6 + 6 = -2x + 12$.

Answer: (D)

10.

MEDIUM · NO CALCULATOR NEEDED

The altitude, in meters, of a drone t seconds after takeoff is modeled by $h(t) = 45 + 12t$ for $0 \leq t \leq 20$. Which of the following is the best interpretation of $h(5) = 105$?

(A) At 5 s, the altitude is 105 m. ✓

(B) The drone climbs 105 m every 5 s.

(C) The drone starts at 105 m.

(D) The drone climbs at 5 m/s.

Worked solution

1. $h(5) = 45 + 12(5) = 105$.
2. The input 5 is a time and the output 105 is an altitude, so the statement describes one moment, not a rate or a starting value.

Answer: (A)

11.

MEDIUM · CALCULATOR

A pool holds 140 liters of water and is filled at a constant rate of 35 liters per minute. What is the least whole number of minutes after which the pool holds at least 1000 liters?

Worked solution

1. Volume after t minutes: $140 + 35t$.
2. $140 + 35t \geq 1000 \Rightarrow t \geq \frac{860}{35} \approx 24.57$.
3. t must be a whole number of minutes, so round *up*: $t = 25$. (Check: $t = 24$ gives $980 < 1000$; $t = 25$ gives 1015.)

Answer: 25

12.

MEDIUM · NO CALCULATOR NEEDED

The points $(1, a)$, $(3, 5)$ and $(7, 11)$ lie on the same line. What is the value of a ?

(A) -2 **(B) 2 ✓**(C) 3 (D) $\frac{7}{2}$ **Worked solution**

1. Slope from $(3, 5)$ to $(7, 11)$: $\frac{6}{4} = \frac{3}{2}$.

2. From $x = 1$ to $x = 3$ the input rises by 2, so y rises by 3: $5 - 3 = a$.

3. $a = 2$.

Answer: (B)

13.

MEDIUM · NO CALCULATOR NEEDED

A line passes through the points $(-3, 5)$ and $(9, -1)$. What is the x -coordinate of its x -intercept?

(A) -7 (B) $\frac{7}{2}$ **(C) 7 ✓**(D) 14 **Worked solution**

1. $m = \frac{-1 - 5}{9 - (-3)} = -\frac{1}{2}$.

2. $y - 5 = -\frac{1}{2}(x + 3) \Rightarrow y = -\frac{1}{2}x + \frac{7}{2}$.

3. Set $y = 0$: $x = 7$.

Answer: (C)

14.

HARD · NO CALCULATOR NEEDED

The function f is linear and $f(t + 6) - f(t) = 21$ for every value of t . If $f(-2) = 3$, what is the value of $f(8)$?

Worked solution

1. A step of 6 in the input adds 21: slope $= \frac{21}{6} = \frac{7}{2}$.

2. From -2 to 8 is a step of 10, so $f(8) = f(-2) + \frac{7}{2} \cdot 10 = 3 + 35 = 38$.

Answer: 38

15.

HARD · NO CALCULATOR NEEDED

If $f(x) = 4x - 3$ and $f(2a) = 3f(a) + 7$, what is the value of a ?

Worked solution

1. $f(2a) = 4(2a) - 3 = 8a - 3$.

2. $3f(a) + 7 = 3(4a - 3) + 7 = 12a - 2$.

3. $8a - 3 = 12a - 2 \Rightarrow -4a = 1 \Rightarrow a = -\frac{1}{4}$.

Answer: $-\frac{1}{4}$

16.

HARD · NO CALCULATOR NEEDED

$$ax + 3y = 12$$

$$4x + 6y = b$$

In the system above, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

Worked solution

1. Infinitely many solutions means both equations describe the same line.
2. Multiply the first equation by 2: $2ax + 6y = 24$. Compare with $4x + 6y = b$: $2a = 4$ and $b = 24$.
3. $a = 2$, $b = 24$, so $a + b = 26$.

Answer: 26

17.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $kx + 3y = 7$ is perpendicular to the line $2x - 5y = 1$, where k is a constant. What is the value of k ?

- (A) $\frac{15}{2}$ ✓
- (B) $\frac{6}{5}$
- (C) $-\frac{15}{2}$
- (D) $\frac{2}{15}$

Worked solution

1. Perpendicular condition: $A_1A_2 + B_1B_2 = 0$.
2. $k(2) + 3(-5) = 0 \Rightarrow 2k = 15$.
3. $k = \frac{15}{2}$.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $Ax + By = C$ is a line, where $A > 0$, $B < 0$ and $C < 0$. Which quadrant does the line *not* pass through?

- (A) Quadrant I
- (B) Quadrant II
- (C) Quadrant III
- (D) Quadrant IV ✓

Worked solution

1. Slope $= -\frac{A}{B} > 0$ (positive).
2. y -intercept $= \frac{C}{B} > 0$ (positive); x -intercept $= \frac{C}{A} < 0$ (negative).
3. A rising line crossing the y -axis above the origin and the x -axis left of the origin passes through Quadrants I, II and III, and never enters Quadrant IV.

Answer: (D)

19.

HARD · NO CALCULATOR NEEDED

Points $A(-2, 9)$ and $B(6, -3)$ are the endpoints of a segment. Which of the following is an equation of the perpendicular bisector of $\overline{AB}$?

(A) $3x - 2y = 0$

(B) $2x + 3y = 13$

(C) $2x - 3y = -5$ ✓

(D) $3x + 2y = 12$

Worked solution

1. Midpoint: $\left(\frac{-2+6}{2}, \frac{9+(-3)}{2}\right) = (2, 3)$.

2. Slope of $\overline{AB}$: $\frac{-3-9}{6-(-2)} = -\frac{3}{2}$; the bisector has slope $\frac{2}{3}$.

3. $y - 3 = \frac{2}{3}(x - 2) \Rightarrow 3y - 9 = 2x - 4 \Rightarrow 2x - 3y = -5$.

4. Check: $3x + 2y = 12$ is line AB itself; $3x - 2y = 0$ has the right point but slope $\frac{3}{2}$ (no negation).

Answer: (C)

20.

HARD · NO CALCULATOR NEEDED

A line with slope -2 passes through the point $(3, 4)$. The line and the positive coordinate axes bound a triangle. What is the area of the triangle?

Worked solution

1. $y - 4 = -2(x - 3) \Rightarrow y = -2x + 10$.

2. Intercepts: $y = 10$; $0 = -2x + 10 \Rightarrow x = 5$.

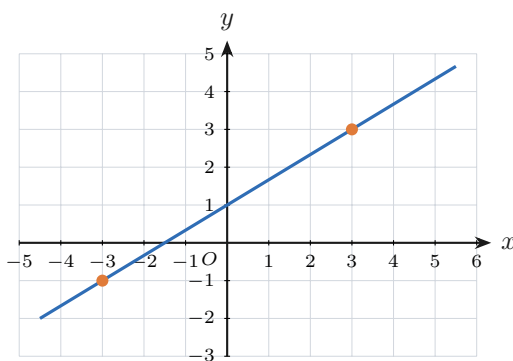
3. Area = $\frac{1}{2}(5)(10) = 25$.

Answer: 25

21.

HARD · DESMOS

The graph of line ℓ is shown. Line m is perpendicular to ℓ and passes through the point $(4, 1)$. What is the y -intercept of line m ?

**Worked solution**

1. Read $(-3, -1)$ and $(3, 3)$ on ℓ : slope $\frac{3 - (-1)}{3 - (-3)} = \frac{2}{3}$.

2. Line m has slope $-\frac{3}{2}$.

3. $1 = -\frac{3}{2}(4) + b \Rightarrow b = 1 + 6 = 7$.

Answer: 7

22.

HARD · CALCULATOR

Company A charges a \$40 setup fee plus \$12 per hour for a rental. Company B charges \$15 per hour with no setup fee. What is the least whole number of hours for which Company A's total cost is less than Company B's?

Worked solution

1. A: $40 + 12h$. B: $15h$.2. $40 + 12h < 15h \Rightarrow 40 < 3h \Rightarrow h > 13.\bar{3}$.3. Whole hours: $h = 13$ gives 196 vs 195 (A is not cheaper); $h = 14$ gives 208 vs 210. The answer is 14.**Answer:** 14

23.

HARD · NO CALCULATOR NEEDED

In an arithmetic sequence, the 4th term is -5 and the 12th term is 27. What is the 20th term?

(A) 55

(B) 59 ✓

(C) 63

(D) 51

Worked solution

1. Common difference: $d = \frac{27 - (-5)}{12 - 4} = 4$.2. $a_{20} = a_{12} + 8d = 27 + 32 = 59$.**Answer:** (B)

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive constant K , the line $3x + 4y = K$ and the two coordinate axes bound a triangle of area 54. What is the value of K ?

Worked solution

1. Intercepts: $x = \frac{K}{3}$ and $y = \frac{K}{4}$.2. Area = $\frac{1}{2} \cdot \frac{K}{3} \cdot \frac{K}{4} = \frac{K^2}{24}$.3. $\frac{K^2}{24} = 54 \Rightarrow K^2 = 1296 \Rightarrow K = 36$.**Answer:** 36

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, line k has slope $\frac{2}{3}$ and passes through $(6, -1)$. Line m is perpendicular to line k and has the same y -intercept as line k . What is the x -coordinate of the x -intercept of line m ?

Worked solution

1. Line k : $-1 = \frac{2}{3}(6) + b \Rightarrow b = -5$.2. Line m has slope $-\frac{3}{2}$ and y -intercept -5 : $y = -\frac{3}{2}x - 5$.3. Set $y = 0$: $-\frac{3}{2}x = 5 \Rightarrow x = -\frac{10}{3}$.**Answer:** $-\frac{10}{3}$

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The line $y = 2x + 4$ crosses the x -axis at point A and the y -axis at point B . Line m passes through A and is perpendicular to $y = 2x + 4$. Line m crosses the y -axis at point C . What is the area of triangle ABC ?

Worked solution

1. $A: 0 = 2x + 4 \Rightarrow A = (-2, 0)$. $B = (0, 4)$.

2. Line m has slope $-\frac{1}{2}$: $y = -\frac{1}{2}(x + 2)$, so $C = (0, -1)$.

3. Base $\overline{BC}$ lies on the y -axis: $BC = 4 - (-1) = 5$. The height is the distance from A to the y -axis, 2.

4. Area $= \frac{1}{2}(5)(2) = 5$.

Answer: 5

27.

SAT CHALLENGE · DESMOS

In the xy -plane, line p is $y = kx + 4$ and line q is $y = (k + 2)x - 6$, where k is a constant. The two lines intersect at a point whose y -coordinate is 19. What is the value of k ?

Worked solution

1. At the intersection: $kx + 4 = (k + 2)x - 6 \Rightarrow 10 = 2x \Rightarrow x = 5$. (k cancels.)

2. On line p : $19 = k(5) + 4 \Rightarrow 5k = 15$.

3. $k = 3$.

Answer: 3

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear functions f and g are defined by $f(x) = 2x + 1$ and $g(x) = ax + b$, where a and b are constants. If $f(g(x)) = 6x - 5$ for all x , what is the value of $g(f(2))$?

Worked solution

1. $f(g(x)) = 2(ax + b) + 1 = 2ax + 2b + 1$.

2. Match $6x - 5$: $2a = 6 \Rightarrow a = 3$; $2b + 1 = -5 \Rightarrow b = -3$, so $g(x) = 3x - 3$.

3. $f(2) = 5$, and $g(5) = 3(5) - 3 = 12$.

Answer: 12

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

A line has x -intercept a and y -intercept b , where $a \neq 0$ and $b = 2a$. If the line passes through $(3, 5)$, what is the value of b ?

Worked solution

1. Intercept form: $\frac{x}{a} + \frac{y}{2a} = 1$. Multiply by $2a$: $2x + y = 2a$.

2. Substitute $(3, 5)$: $2(3) + 5 = 11 = 2a$.

3. $b = 2a = 11$.

Answer: 11

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

For linear function p , $p(c) = -6$, $p(5) = 42$, and the slope of the graph of p is 8. For linear function t , $t(c) = -7$ and $t(6) = 56$. What is the slope of the graph of t ?

Worked solution

1. For p : $\frac{42 - (-6)}{5 - c} = 8 \Rightarrow 5 - c = 6 \Rightarrow c = -1$.

2. For t : the points are $(-1, -7)$ and $(6, 56)$.

3. Slope of t : $\frac{56 - (-7)}{6 - (-1)} = \frac{63}{7} = 9$.

Answer: 9